AMIRIS Germany2027 - Build Documentation

Every input's origin and validation, and the full record of what was copied, edited, or newly written across every YAML file

Muideen Oladayo Ajiboye | 13 August 2026

Parts 1-6 cover the first built version of the Germany2027 scenario: no cross-border trade (gap #5, assumed zero), and a single BDEW-for-everything demand profile (V1). Parts 7-9 extend this with the two later builds: the component-split demand profile (V2) and the with-import cross-border variant, and give the full four-way comparison across every combination now built (V1/V2 demand x no-import/with-import).

How to Read the Source Labels Used Throughout

Label	Meaning
Brainpool (direct)	Value taken straight from Amiris_Inputdata_EN.xlsx with no conversion.
Brainpool (converted)	Value taken from Brainpool's data but transformed - unit conversion, regional averaging, or capacity-weighted blending - before use.
AMIRIS own	Brainpool supplied nothing for this field. AMIRIS's own validated Germany2019 example scenario value was reused as the gap-fill, per the earlier documented data-gaps decision.
Researched (external)	Neither Brainpool nor AMIRIS's own data covered this. A real external source was looked up this session (EEG/BNetzA auction rates, MaStR battery statistics, renewables.ninja cross-check).
Placeholder (flagged gap)	No real figure exists anywhere yet (confirmed by research, not assumed). AMIRIS's own example value is used as the least-bad stand-in, explicitly flagged as unresolved.

Part 1 - Timeseries Inputs (the timeseries/ folder)

A. Fuel and carbon prices

File	Source	Method
hard_coal_price.csv	Brainpool (converted)	USD/tonne to EUR/MWh(thermal): divide by the file's own monthly USD/EUR rate, then by 8.141 MWh/tonne (standard hard-coal calorific equivalent, ~29.3 GJ/t).
oil_price.csv	Brainpool (converted)	USD/barrel to EUR/MWh(thermal): divide by the monthly USD/EUR rate, then by 1.70 MWh/barrel (~5.8 MMBtu/barrel, standard Brent equivalent).
natural_gas_price.csv	Brainpool (direct)	'Natural Gas Germany [EUR/MWh]' column used as-is - already the correct unit.
co2_price.csv	Brainpool (direct)	'EUA - EU Emission Allowance [EUR/tCO2]' column used as-is - already AMIRIS's expected unit.
Lignite fuel price (in MarketsAndForecast.yaml, not a file)	AMIRIS own	Flat 5.00 EUR/MWh. Brainpool's Variable_cost sheet has no lignite series at all - realistic, since lignite is domestically mined and not internationally traded like hard coal/gas/oil.
Nuclear fuel price	N/A	Omitted entirely - 0 GW nuclear capacity in Brainpool's 2027 figures (post phase-out).

Format note (verified before building, not assumed):

AMIRIS's own fuel-price files use monthly timestamps with fameio interpolating between them - confirmed by inspecting hard_coal_price.csv in the existing Germany2019 scenario before writing any conversion code. This meant Brainpool's monthly data could be written directly into the same format with no resampling needed.

B. Outage and must-run factors

Files	Source	Method
lignite_outage.csv, lignite_must_run.csv, hard_coal_outage.csv, hard_coal_must_run.csv, natural_gas_outage.csv, natural_gas_must_run.csv	AMIRIS own	Brainpool supplies no outage data (correctly - unplanned 2027 outages cannot be known in 2026). AMIRIS's own validated 2015-2019 files reused, with every timestamp's calendar year shifted (2019->2027, 2020->2028, 2021->2029, later also 2018->2026 once that edge case was found - see Part 3), values and month/day/time otherwise untouched.
Oil outage	AMIRIS own	A flat 0.07 (7%) figure, not a file - matches AMIRIS's own Germany2019 convention for this fuel type.

C. Renewable generation profiles

File	Source	Method
solar_openfield_profile.csv	Brainpool (converted)	Regional average of the four PV_openfield_[north/east/middle/swest] columns in the feedinprofile sheet, per hour. Feeds the Grid Feed-in solar agent.
solar_rooftop_profile.csv	Brainpool (converted)	Regional average of the four PV_rooftop_[region] columns. Feeds the Prosuming solar agent. Kept separate from openfield rather than blended into one profile - see Part 4.
wind_onshore_profile.csv	Brainpool (converted)	Regional average of the four Wind_[region] columns.
wind_offshore_profile.csv	AMIRIS own, validated	Brainpool gives capacity but no offshore-specific shape (only generic Wind_[region] columns, used for onshore). AMIRIS's own profile reused, redated to 2027. Validated: implies a 37.1% annual capacity factor, which sits inside the real German offshore fleet's 37-45% range (EnergyNumbers.info, Fraunhofer research this session).
run_of_river_profile.csv	AMIRIS own	Brainpool gives a Run-of-River capacity figure but no hourly shape. AMIRIS's own profile reused, redated to 2027.
other_res_profile.csv	AMIRIS own	Built but not currently referenced by any 2027 agent - Brainpool's 'Other Renewables' was mapped entirely onto the Biomass profile instead (see Part 4). Kept on disk for a possible future split.
biomass_profile.csv	AMIRIS own	Brainpool gives an 'Other Renewables' capacity figure but no dispatch shape. AMIRIS's own FROM_FILE dispatch profile reused, redated to 2027.

D. Demand (load_v1_bdew.csv)

Component	Value	Source
Inflexible base demand	604.85 TWh	Brainpool (direct)
Electrolysis (hydrogen)	14.97 TWh	Brainpool (direct)
Electric mobility (EV charging)	17.81 TWh	Brainpool (direct)
Heat pumps	18.69 TWh	Brainpool (direct)
Combined total used	**656.32 TWh**	**Sum of the four rows above**

Method: demandlib's BDEW ElecSlp.get_scaled_power_profiles({'h0': 656,320,000 MWh}) - the real BDEW Standardlastprofil methodology (12 representative daily curves by season and day-type, H0's day-of-year dynamization applied automatically), at native 15-minute resolution, resampled to hourly to match AMIRIS's expected input. A round-trip check confirmed the built file's hourly values sum back to exactly 656.32 TWh.

Two figures deliberately excluded from the total:

Net Exports (a cross-border trade balance, out of scope for this no-import build per gap #5) and Pumped Storage Losses (already implicit in the storage agents' own charging/discharging efficiency in Storage.yaml - adding it separately would double-count).

Part 2 - Conventional Fleet (Conventionals.yaml)

Field	Lignite	Hard Coal	Nat. Gas	Oil + Other Fossil
Capacity (MW)	13,927 (Brainpool)	5,504 (Brainpool)	36,237 (Brainpool)	9,790 (Brainpool: 1,540 Oil + 8,250 Other Fossil, merged)
Block size (MW)	500 (AMIRIS own)	300 (AMIRIS own)	200 (AMIRIS own)	100 (AMIRIS own)
Efficiency range	0.3108-0.45 (AMIRIS own)	0.339-0.492 (AMIRIS own)	0.516-0.617 (AMIRIS own CCGT range, used as representative)	0.311-0.397 (AMIRIS own)
Variable opex (EUR/MWh)	2.0 (AMIRIS own)	2.5 (AMIRIS own)	1.2 (AMIRIS own)	1.2 (AMIRIS own)
Markup band (EUR/MWh)	-60 to 0 (AMIRIS own)	-15 to 5 (AMIRIS own)	-10 to 10 (AMIRIS own CCGT band)	0 to 0 (AMIRIS own)
CO2 factor (t/MWh)	0.407 (Brainpool)	0.338 (Brainpool)	0.202 (Brainpool)	0.266 (Brainpool Oil factor, used as proxy for the merged category)

New gap surfaced while building (not in the original 5-item list):

Brainpool's Variable_cost sheet covers fuel commodity prices, not per-fuel-type variable OPERATING cost (a separate concept - non-fuel O&M per MWh). AMIRIS's own opex figures were carried over for all four categories, consistent with the established 'AMIRIS fills gaps Brainpool doesn't cover' approach, but this specific gap had not been explicitly flagged before now.

Two simplifications, both documented rather than silently made: natural gas is modelled as ONE combined block (CCGT+OCGT merged) since Brainpool gives only one gas capacity figure with no technology split; and Oil is merged with 'Other Fossil' into a single AMIRIS Oil-type block, consistent with AMIRIS's own scenario comment that its Oil category already covers 'oil, other fossil fuels, mixed fossil fuels.'

Part 3 - Renewable Fleet and Subsidies (RenewablesAndPolicy.yaml)

Agent	Capacity	Support mech	Reference rate	Rate source
Solar Openfield (Grid Feed-in)	119,179 MW (Brainpool)	MPVAR	49 EUR/MWh	Researched: BNetzA solar auction, March 2026 round (4.94 ct/kWh volume-weighted award).
Solar Rooftop (Prosuming)	33,706 MW (Brainpool)	FIT	75 EUR/MWh	Researched: EEG Section 49 degression schedule extrapolated to 2027. FLAGGED: draft EEG-Novelle 2026 (cabinet-approved 29 Jul 2026, not yet law) proposes ending this support for new PV <25kW from 2027.
Wind Onshore	90,232 MW (Brainpool)	MPVAR	52 EUR/MWh	Researched: BNetzA wind-onshore auctions, Feb-Aug 2026 rounds (5.06-5.54 ct/kWh), midpoint.
Wind Offshore	12,573 MW (Brainpool)	MPVAR	187 EUR/MWh	PLACEHOLDER (flagged gap): no real 2027 figure exists anywhere - BNetzA's 2025 offshore auctions received zero bids and the next round is postponed to 2027 itself. AMIRIS's own Germany2019 offshore clusters' capacity-weighted LCOE used as the least-bad stand-in.
Run-of-River	4,161 MW (Brainpool)	FIT	100 EUR/MWh	AMIRIS own - no EEG hydro-specific rate was found; AMIRIS's own Germany2019 value reused.
Biomass (proxy for 'Other Renewables')	8,250 MW (Brainpool)	MPVAR	197 EUR/MWh	Researched: BNetzA biomass tender, April 2026 round, cleared near the published ceiling (19.43-19.83 ct/kWh, undersubscribed).

Design correction made mid-build:

Solar was originally planned as one blended agent (per an earlier decision to sum Grid Feed-in and Prosuming totals). This had to be reversed: AMIRIS assigns one SupportInstrument per operator agent, and Grid Feed-in (MPVAR) and Prosuming (FIT) use genuinely different mechanisms, so they cannot share one agent. Kept as two agents instead, each with its own real capacity and profile - which mirrors exactly how AMIRIS's own Germany2019 scenario handles multiple solar sub-fleets under *one shared trader.*

Deliberate departure from AMIRIS's own example:

AMIRIS's Germany2019 scenario leaves biomass completely unsupported (routed through NoSupportTrader). This build instead gives it a real EEG-researched MPVAR rate (197 EUR/MWh), since that is a more accurate reflection of how German biomass is actually subsidised - a conscious improvement on AMIRIS's own placeholder, not an oversight.

Reservoir Hydro (1,540 MW, Brainpool) is NOT in this file - it is modelled as a dispatchable storage agent instead of a weather-driven renewable, since reservoir hydro output is an operator choice (when to release water), unlike run-of-river (flow-determined). See Part 4.

'Other Renewables' (8,250 MW, Brainpool) was mapped entirely onto the Biomass agent rather than split further, since biomass is the dominant real-world component of Germany's 'sonstige Erneuerbare' statistical bucket - a documented simplification, not a literal Brainpool breakdown.

Part 4 - Storage Fleet (Storage.yaml)

Agent	Power (Brainpoo	Duration	Energy capacity	Efficiency (charge/discharge)
Pumped Hydro (Pumpspeicher)	8,377.6 MW	6.399 h - AMIRIS own, capacity-weighted average across AMIRIS's own 16 daily-cycle Germany2019 storage units	53,607.5 MWh	87.5% / 85.2% - same AMIRIS-own fleet average
Large-Scale Battery	5,151.7 MW	2.0 h - Researched (MaStR-derived): RWE's Hambach project, 236 MW / 470 MWh, targets 2027 commissioning - the same year as this scenario	10,303.4 MWh	95% / 95% - standard Li-ion round-trip assumption (no German-specific efficiency statistic was found; duration was researched, efficiency was not)
Reservoir Hydro	1,540.0 MW	111.22 h - AMIRIS own, borrowed from the long-duration outlier unit originally excluded from the pumped-hydro duration calculation - understood, on reflection, to represent exactly this kind of seasonal Alpine reservoir	171,282.2 MWh	91% / 91% - same AMIRIS-own unit

MaStR access itself: the Marktstammdatenregister's bulk XML download and web-search CSV export are public with no login required; only its automated API needs registration (same low-friction pattern as the ENTSO-E token obtained earlier in this project). The battery duration figure was sourced from a third party (Modo Energy) that aggregates MaStR project-level data, not from a raw MaStR pull.

Part 5 - YAML File Construction: What Was Copied, Edited, or Written New

File	Status	What was actually done
schema.yaml	Copied unchanged	Generic AMIRIS agent-type schema, identical across all scenarios - no scenario-specific data, so copied directly from Germany2019 with no edits.
scenario.yaml	Written new	New Metadata block (runId 'Germany2027', description noting the data provenance and no-import assumption). Simulation window set to 2026-12-31_23:58:00 to 2027-12-31_23:58:00. The PolicySet StringSet enum was replaced with this scenario's six actual subsidy scheme names (SolarRooftopFit, RunOfRiverFit, SolarOpenfieldMpvar, WindOnMpvar, WindOffMpvar, BiomassMpvar) - AMIRIS validates PolicySet values against this list, so Germany2019's own names would not have matched.
agents/Conventionals.yaml	Written new	Four fuel-type blocks (Lignite, Hard Coal, Natural Gas, Oil+Other Fossil) built from the Part 2 table above. Nuclear's builder/trader/operator triad and the separate OCGT block were both omitted entirely, rather than left at zero, since AMIRIS has no real use for a zero-capacity plant builder.

File	Status	What was actually done
agents/RenewablesAndPolicy.yaml	Written new	Six VariableRenewableOperator/Biogas agents, one shared RenewableTrader (MPVAR) and one shared SystemOperatorTrader (FIT), and a SupportPolicy agent carrying the six researched/gap-filled reference rates from Part 3.
agents/Storage.yaml	Written new	Three GenericFlexibilityTrader agents (Pumpspeicher, Battery, Reservoir Hydro) built from the Part 4 table. Pumpspeicher and Reservoir use MIN_SYSTEM_COST/ENSURE_DISPATCH (matching AMIRIS's own convention for large legacy hydro); Battery uses MAX_PROFIT/STORAGE_CONTENT_VALUE (matching AMIRIS's own convention for merchant-style assets).
agents/Demand.yaml	Written new	One DemandTrader pointing at load_v1_bdew.csv, ValueOfLostLoad kept at AMIRIS's own 3,000 EUR/MWh (the price ceiling seen whenever the market cannot fully meet demand).
agents/MarketsAndForecast.yaml	Written new	DayAheadMarketSingleZone, CarbonMarket, FuelsMarket, and SensitivityForecaster agents, wired to the Part 1 timeseries files. Structurally identical to Germany2019's version except the fuel-price file references and the removal of the Nuclear fuel-price entry.

Contract file	Status	What was actually done
contracts/conventionals.yaml	Copied, then edited	AgentGroups builder/trader/operator ID lists trimmed to [2001,2002,2003,2005] / [1001,1002,1003,1005] / [501,502,503,505] - dropping the Nuclear (2000/1000/500) and OCGT (2004/1004/504) slots. All timing/product-name contract logic below the AgentGroups block is generic and was left untouched.
contracts/demand.yaml	Copied unchanged	References only Id 100 (DemandTrader), which is identical in this build - no edit needed.
contracts/renewables_renewableTrader.yaml	Copied, then edited	Renewables ID list trimmed to [60, 70, 80, 52] - the four MPVAR-marketed agents (Solar Openfield, Wind Onshore, Wind Offshore, Biomass).
contracts/renewables_systemOperator.yaml	Copied, then edited	Renewables ID list trimmed to [61, 50] - the two FIT-marketed agents (Solar Rooftop, Run-of-River).
contracts/renewables_noSupport.yaml	NOT created	Germany2019's version routes Biogas/Other through the NoSupportTrader. This build gives every renewable category a real subsidy mechanism (see Part 3), so nothing routes through NoSupportTrader (Id 12) at all - the file, and the agent, were both omitted.
contracts/storage.yaml	Copied, then edited	Storage ID list trimmed from Germany2019's eighteen units down to this build's three: [700, 701, 702].
contracts/supportPolicy.yaml	Copied, then edited	allMarketers trimmed from [11, 12, 13] to [11, 13], matching the removal of NoSupportTrader above.

Agent ID numbering scheme

Reused Germany2019's own per-category ID ranges wherever a directly matching category existed (builders 2000s, traders 1000s, conventional operators 500s, renewable operators 50-83, storage 700s, market/system agents 1/3/4/6, demand 100), simply omitting the slots for categories this build doesn't use (Nuclear, OCGT, NoSupportTrader) rather than renumbering everything from scratch. This kept the trimmed contract files a close, easily-checked diff against the originals instead of a full rewrite.

Part 6 - Validation and QA Performed

Before writing any code

Inspected AMIRIS's own existing timeseries files (hard_coal_price.csv, co2_price.csv, nuclear_outage.csv, load.csv, scenario.yaml, every agents/*.yaml and contracts/*.yaml file) to confirm the exact timestamp format, delimiter, and resolution fameio expects, rather than assuming a format and risking a rebuild later.

Structural validation

Ran the scenario through amirispy's actual compile step (amiris run), which converts the YAML into a protobuf input file before the Java engine executes it - this surfaces structural/schema errors within seconds, before committing to a multi-minute full simulation.

Bug 1 found and fixed: missing SupportInstrument on the Biogas agent

```
Exception in thread "main" java.lang.RuntimeException: BiomassMpvar: Policy not
configured for instrument null
    at agents.policy.SetPolicies.register(SetPolicies.java:64)
```

Cause: AMIRIS's own Germany2019 Biogas entry never needed a SupportInstrument attribute, because it routes through NoSupportTrader (no subsidy). This build gives Biogas a real MPVAR subsidy (Part 3), which requires that attribute to be explicit - it was missing from the first draft. Fixed by adding 'SupportInstrument: MPVAR' to the agent.

Bug 2 found and fixed: four renewable profiles never redated to 2027

The scenario compiled and ran successfully after Bug 1's fix, but the result was not physically plausible: 25.3% of all 8,760 hours cleared at the 3,000 EUR/MWh shortage price ceiling. Diagnosis, not guesswork: picked a specific shortage hour (2027-01-16 18:00) and checked every agent's actual award at that hour. Total awarded energy across conventional, renewable, and storage agents matched the load served exactly - ruling out a bidding/market-clearing bug and confirming a genuine supply shortfall. Wind Offshore, Run-of-River, and Biomass all showed exactly zero at that hour despite no physical reason to be zero simultaneously; checking their source CSVs directly showed the cause: wind_offshore_profile.csv, run_of_river_profile.csv, other_res_profile.csv, and biomass_profile.csv had been copied from Germany2019 with their original 2019 timestamps intact, never redated to 2027 like the outage files were - so their data never overlapped this scenario's simulated calendar year at all. Fixed by applying the same redate function already used for the outage files to these four profiles as well.

Result after the fix:

Shortage hours fell from 25.3% to 15.9% of the year (2,216 to 1,391 hours), and negative prices appeared for the first time (108 hours) - a healthy sign, since AMIRIS's own example scenarios structurally cannot produce negative prices at all (see the 2018/2019 backtest root-cause findings earlier in this project).

Remaining shortage pattern: diagnosed as expected V1 behaviour, not a bug

Broke the remaining 1,391 shortage hours down by month and hour-of-day. 71% (995 hours) fall in the 18:00-21:00 evening window, worst in December and August. This is the textbook signature of a household evening demand spike - consistent with V1's known, deliberate simplification of applying one BDEW household (H0) load shape to the entire economy's demand, including industrial load, EV charging, and heat pumps, none of which actually follow a household's evening ramp. This is reported as a finding for the V2 (component-split) build to address, not chased further as a bug in this V1 build.

Part 7 - Demand Profile V2 (Component-Split Build)

V1 applied a single BDEW household (H0) Standardlastprofil to the entire 656.32 TWh combined demand total. V2 replaces this with a purpose-built shape for each of Brainpool's four demand components separately, then sums them (build_2027_demand_v2_split.py).

Component	Value	Method
Inflexible base	604.85 TWh	Real 2023 ENTSO-E national load shape (examples/backtest/Germany2023/timeseries/load.csv, pulled this project from the ENTSO-E Transparency Platform), rescaled hour-by-hour so the annual total matches the 2027 target. Keeps the real seasonal/hour-of-day pattern; day-of-week alignment between 2023 and 2027 is not preserved (a documented simplification).

Component	Value	Method
Heat pumps	18.69 TWh	demandlib's BDEW HeatBuilding profile (temperature-driven Standardlastprofil methodology), driven by real DWD Test Reference Year climate data bundled with demandlib itself, averaged across all 15 German climate regions for a national series (mean 8.5C, range -3.9C to 24.9C). Simplification: treats heat-pump electrical demand as directly proportional to heat demand, i.e. assumes a constant coefficient-of-performance (COP) across the year rather than modelling the real efficiency drop in cold weather.
Electrolysis	14.97 TWh	Flat, constant hourly profile (1,708 MW every hour) - industrial electrolysers are typically run close to constant for efficiency.
E-mobility	17.81 TWh	An assumed, documented daily charging shape - evening-peaked (weights rising through the afternoon to a peak around 18:00, matching typical unmanaged/home-dominant EV charging patterns), repeated identically every day. No seasonal or weekday/weekend variation is modelled - the least standardised of the four components, since no real German EV charging load-curve dataset was available to reshape instead.

Round-trip check:

Combined total = 656.3172 TWh against a 656.3167 TWh target - matches to within rounding. Combined peak load = 106,178.7 MW, versus V1's 138,146.7 MW and Brainpool's own stated Annual Peak Load of 119,001 MW - V2 sits much closer to the real target, though still below it, since the four components' individual peaks do not all coincide in the same hour.

Result: V2 materially reduces V1's shortage problem

Metric	V1 (BDEW-for-everything)	V2 (component-split)
Peak load	138,146.7 MW	106,178.7 MW
Shortage hours (>=3000 EUR/MWh)	1,391 (15.88%)	659 (7.52%)
Negative-price hours	108 (1.23%)	884 (10.09%)
Mean price	519.02 EUR/MWh	285.58 EUR/MWh
Median price	67.52 EUR/MWh	77.19 EUR/MWh

This confirms the diagnosis made at the end of Part 6: V1's shortage hours were concentrated in the 18:00-21:00 evening window because a household evening-peak shape had been applied to the whole economy's demand. Spreading the heat-pump, electrolysis, and EV components across their own, less peaky shapes removes most of that artificial peak, cutting shortage hours by more than half and mean price by 45%. The new build (examples/backtest/Germany2027_DemandV2/, referencing load_v2_split.csv) compiled and ran with the same zero-error validation as V1.

Part 8 - With-Import Cross-Border Variant (ImportTrader)

Gap #5 assumed zero cross-border trade for the first build. This variant captures it: Brainpool's Capacity sheet gives 'Net Exports' = -43.90 TWh for 2027, i.e. Germany is projected to be a NET IMPORTER of 43.90 TWh - the same direction as the real 2023 outturn, where Germany was a net importer for the first time in decades (24.4 TWh net, computed from real ENTSO-E cross-border flow data pulled for this build).

Scope limitation, by design:

AMIRIS's ImportTrader agent type (schema.yaml, and AMIRIS's own SimpleCoupled demo scenario, used as the real wiring template) can only ADD supply to the market - offering imported energy as an extra bid. There is no ExportTrader for a single, uncoupled market zone, so this variant captures only the import side of cross-border trade. Modelling Germany's real export flows as well would require a full multi-zone MarketCoupling setup, which was judged out of scope for this build.

Attribute	Method
AvailableEnergyForImport	Real 2023 net cross-border flow (total physical imports minus total physical exports, both pulled per-hour for DE_LU from the ENTSO-E Transparency Platform, summed across all 11 neighbouring zones: AT, BE, CH, CZ, DK_1, DK_2, FR, NO_2, NL, PL, SE_4), clipped to zero (only the 5,014 hours Germany was a real net importer are usable), then rescaled so the annual total matches Brainpool's 43.90 TWh 2027 target. Preserves the real historical timing of import need rather than inventing a shape.
ImportCostInEURperMWH	Real 2023 French day-ahead price (mean 96.86, min -134.94, max 276.12 EUR/MWh), used as a proxy for import cost. France is Germany's largest, single most stable interconnector partner (large, steady nuclear baseload) - a documented simplification standing in for a flow-weighted mix of all 11 neighbouring price zones.

Finding: realized import volume falls well short of the offered 43.90 TWh

AMIRIS's ImportTrader is a genuine price-competing bid, not a forced quantity: it only clears in hours where the German market price would otherwise exceed the French price offered that hour. The market only actually dispatched a fraction of what was offered:

Build	Offered	Awarded (cleared)	Utilization
V1 demand + import	43.90 TWh	12.33 TWh	28.09%
V2 demand + import	43.90 TWh	7.40 TWh	16.87%

Reported as a finding, not a bug (explicit decision):

This gap demonstrates a real, meaningful divergence between Brainpool's fixed net-import ASSUMPTION (a top-down forecaster's exogenous balance figure) and AMIRIS's endogenous, price-competitive DISPATCH of that same import capacity (an agent-based market clearing outcome). V1's peakier demand shape pushes German prices above the French benchmark more often, so more of the offered import volume clears there (28.09%) than in V2 (16.87%), even though V2 is otherwise the more realistic demand build. This asymmetry is itself a useful methodological data point for the thesis's central AMIRIS-vs-Brainpool comparison.

Market impact of the imports that do clear

Metric	V1 no-import	V1 with-import	V2 no-import	V2 with-import
Shortage hours	15.88%	11.78%	7.52%	5.35%
Negative-price hours	1.23%	1.68%	10.09%	14.60%
Mean price (EUR/MWh)	519.02	400.04	285.58	219.81

Even the partial import volume that clears does real work: it relieves scarcity (shortage hours fall in both demand versions) and pulls the mean price down materially (23-32%). Negative-price hours rise slightly with imports in both cases, since the French price series itself goes negative in some hours (min -134.94 EUR/MWh), occasionally adding cheap/negative-priced import supply on top of an already renewables-saturated German hour.

New scenario folders: examples/backtest/Germany2027_WithImport/ (V2 demand + import) and examples/backtest/Germany2027_V1WithImport/ (V1 demand + import), both compiled and ran with zero errors, agent Id 112, wired per AMIRIS's own SimpleCoupled demo contract pattern.

Part 9 - Four-Way Version Matrix: Full Comparison

All four combinations of the originally planned 2x2 build matrix (demand V1/V2 x no-import/with-import) are now built, compiled, and simulated end to end.

Build	Peak load	Shortage hrs	Negative hrs	Mean price	Import cleared
V1, no-import	138,146.7 MW	15.88%	1.23%	519.02	n/a
V1, with-import	138,146.7 MW	11.78%	1.68%	400.04	12.33 / 43.90 TWh
V2, no-import	106,178.7 MW	7.52%	10.09%	285.58	n/a
V2, with-import	106,178.7 MW	5.35%	14.60%	219.81	7.40 / 43.90 TWh

Reading across the matrix: demand shape (V1 vs V2) is the dominant driver of shortage relief - moving from V1 to V2 alone cuts shortage hours from 15.88% to 7.52% (no-import) and from 11.78% to 5.35% (with-import). Cross-border imports are a smaller, second-order effect on top of that, but still meaningfully reduce both shortage frequency and mean price within either demand version. Folder locations: Germany2027 (V1 no-import), Germany2027_DemandV2 (V2 no-import), Germany2027_WithImport (V2 with-import), Germany2027_V1WithImport (V1 with-import).

Part 10 - Complete List of Open Flags Across All Builds

#	Flag	Where it lives
1	Wind offshore subsidy rate (187 EUR/MWh) is an AMIRIS-own placeholder - no real 2027 figure exists anywhere yet, confirmed by research, not assumed.	RenewablesAndPolicy.yaml, SupportPolicy WindOffMpvar
2	Solar rooftop FIT (75 EUR/MWh) assumes current EEG law continues; a draft 2026 reform (not yet passed) would end this support for new installations <25kW from 2027.	RenewablesAndPolicy.yaml, SupportPolicy SolarRooftopFit
3	Natural gas modelled as one combined CCGT+OCGT block - Brainpool gives no technology split.	Conventionals.yaml
4	Oil and Other Fossil merged into one AMIRIS Oil-type block; the combined CO2 factor uses Oil's own value as a proxy, since Brainpool gives no separate 'Other Fossil' factor.	Conventionals.yaml
5	Variable operating cost (opex) for all four conventional categories uses AMIRIS's own values - Brainpool's Variable_cost sheet covers commodity fuel prices, not plant O&M cost, and this gap was not in the original 5-item gap list.	Conventionals.yaml
6	Battery efficiency (95%/95%) is a standard Li-ion assumption, not a researched German-specific figure - only duration was independently researched (MaStR/RWE Hambach).	Storage.yaml
7	'Other Renewables' assumed to be entirely biomass, with no split for the minor geothermal/other component AMIRIS's own schema would otherwise support separately.	RenewablesAndPolicy.yaml
8	RESOLVED by V2: V1's demand shape produced a peak (138.1 GW) well above Brainpool's stated Annual Peak Load (119.0 GW). V2's component-split build closes most of this gap (106.2 GW) - confirmed by simulation, not just expected.	load_v1_bdew.csv vs load_v2_split.csv
9	Heat-pump profile (V2) assumes a constant coefficient-of-performance (COP) across the year - does not model the real efficiency drop in cold weather, which would make the electrical profile even peakier in winter than modelled.	build_2027_demand_v2_split.py
10	E-mobility profile (V2) is an assumed daily shape, not sourced from real EV charging data, with no seasonal or weekday/weekend variation - the least standardised of the four V2 demand components.	build_2027_demand_v2_split.py
11	Import cost uses a single-neighbour proxy (real French day-ahead price), not a flow-weighted mix of all 11 of DE_LU's actual neighbouring price zones.	build_2027_import.py, ImportCostInEURperMWH.csv
12	With-import builds model only the import side of cross-border trade - no ExportTrader exists for a single, uncoupled market zone, so Germany's real export flows are not represented at all.	agents/Import.yaml
13	Realized import volume (7.40-12.33 TWh) falls well short of the offered/Brainpool-target 43.90 TWh, since AMIRIS's ImportTrader is a genuine price-competing bid rather than a forced quantity. Reported as a deliberate methodological finding, not corrected to force full dispatch.	ImportTrader.csv result output